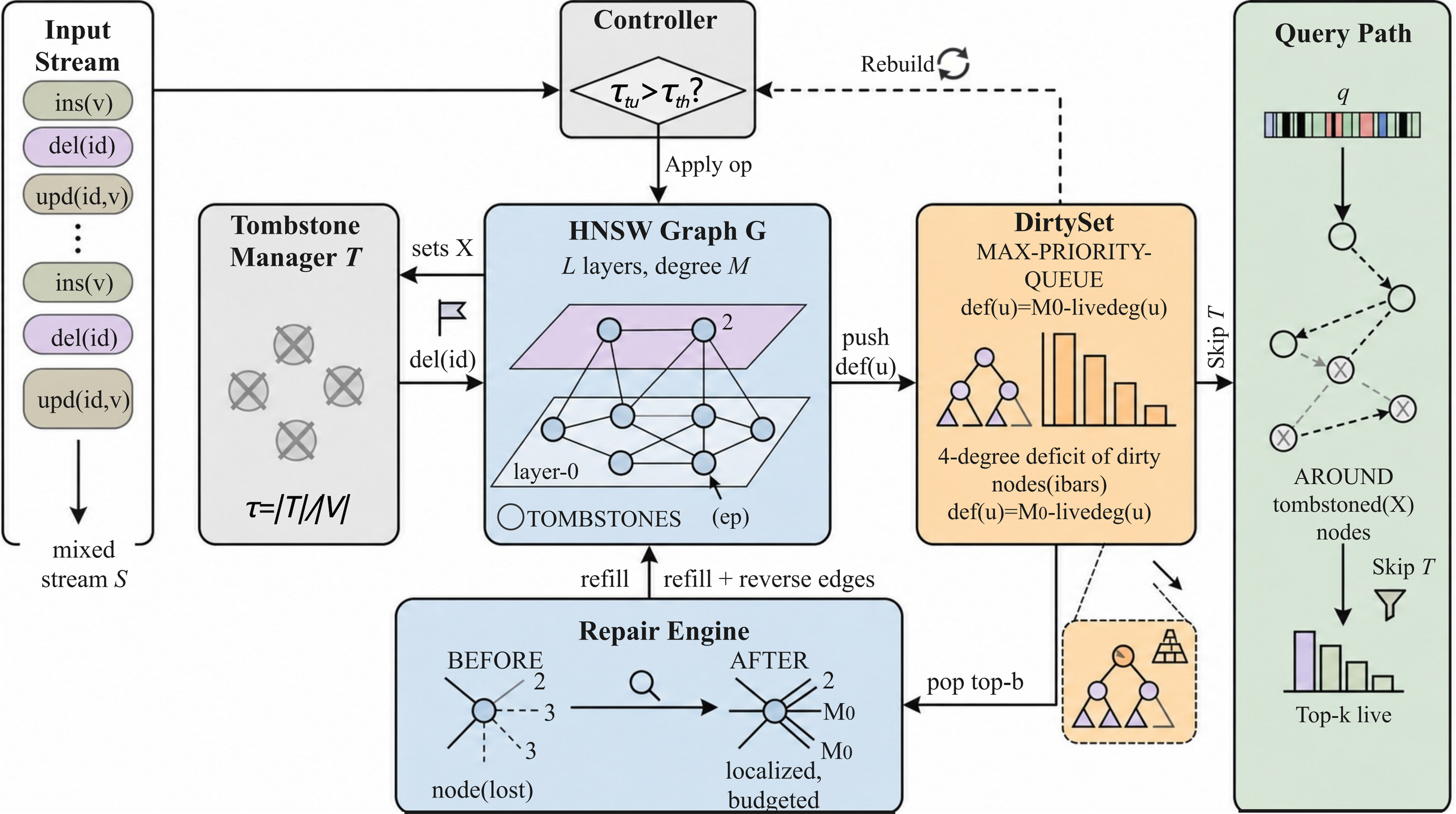

Input Stream
ins(v)
del(id)
upd(id,v)
ins(v)
del(id)
upd(id,v)
mixed stream S
Controller
τtu>τth?
Rebuild
Apply op
Tombstone Manager T
τ=|T|/|V|
sets X
del(id)
HNSW Graph G
L layers, degree M
2
layer-0
TOMBSTONES
(ep)
push def(u)
DirtySet
MAX-PRIORITY-QUEUE
def(u)=M0-livedeg(u)
4-degree deficit of dirty nodes(ibars)
def(u)=M0-livedeg(u)
Skip T
Query Path
q
AROUND tombstoned(X) nodes
Skip T
Top-k live
refill
refill + reverse edges
Repair Engine
BEFORE
2
3
3
node(lost)
AFTER
2
M0
M0
localized, budgeted
pop top-b